BT Practical 15: Installation of MetaMask and Study of Ether Spent per Transaction

Type: No code. Practical work in the browser; submit screenshots plus a short write-up.

What you need

- Google Chrome (or Brave/Edge/Firefox)
- Internet connection
- About 20 minutes

Part A: Install MetaMask

1. Open **https://metamask.io** and click **Download**. Use only the official site; fake extensions steal funds.
2. Click **Add to Chrome**, then **Add extension**.
3. Pin it: click the puzzle icon (Extensions) in the Chrome toolbar and click the pin next to MetaMask.

[Screenshot] *Screenshot 1: MetaMask fox icon in the toolbar.*

4. Click the fox icon. Tick "I agree to the Terms of use", then click **Create a new wallet**.
5. Choose whether to share usage data (either option is fine).
6. Create a **password** (8+ characters). It unlocks MetaMask on this computer only.
7. Click **Secure my wallet**, then **Reveal Secret Recovery Phrase**. Write the 12 words on paper, in order.

Never share them and never screenshot them.

8. Confirm the missing words it asks for, then click **Got it / Done**.

[Screenshot] *Screenshot 2: MetaMask home screen showing "Account 1" and 0 ETH (hide your address if you like).*

Part B: Switch to a test network

9. Click the **network dropdown** (top-left, shows "Ethereum Mainnet").
10. Turn on **Show test networks** (or go to Settings → Advanced → Show test networks → ON).
11. Select **Sepolia** (the main Ethereum test network).

[Screenshot] *Screenshot 3: Sepolia selected.*

Part C: Get free test Ether (faucet)

12. Click your account name to **copy your address** (starts with `0x...`).
13. Open a Sepolia faucet, for example:
    - https://cloud.google.com/application/web3/faucet/ethereum/sepolia
    - https://www.alchemy.com/faucets/ethereum-sepolia
    - https://sepolia-faucet.pk910.de (PoW faucet: mine in the browser for a few minutes)
14. Paste your address, solve the captcha and click **Send / Request**.
15. Wait 10–60 seconds. MetaMask shows a balance like **0.05 SepoliaETH**.

[Screenshot] *Screenshot 4: Balance received.*

Part D: Send a transaction and study the gas fee

16. Create a second account: account dropdown → **Add account** → "Account 2". Copy its address.
17. Switch back to **Account 1**, then click **Send**.
18. Paste Account 2's address and enter the amount **0.001**.
19. Before confirming, look at the **Network fee / Estimated fee** section. Click **Edit / Market** to see:

- **Gas limit**: 21000 for a simple transfer
- **Base fee** (Gwei)
- **Priority fee / tip** (Gwei)
- **Max fee**

[Screenshot] *Screenshot 5: The fee details before confirming.*

20. Click **Confirm**.
21. Open the **Activity** tab, click the transaction, then **View on block explorer** (Etherscan).

[Screenshot] *Screenshot 6: Etherscan page.*

22. On Etherscan, note **Gas Used**, **Gas Price**, **Transaction Fee** and **Block number**.
23. Repeat steps 17–22 with a **Low** and an **Aggressive** gas setting (MetaMask's "Edit" fee menu).

Compare the fees and confirmation times.

Part E: Fill this observation table (include it in your write-up)

#	Amount sent	Gas limit	Gas used	Gas price (Gwei)	Fee (ETH)	Time to confirm
1	0.001 ETH	21000	21000	____	____	____
2	0.001 ETH (Low)	21000	21000	____	____	____
3	0.001 ETH (Aggressive)	21000	21000	____	____	____

Formula: Transaction fee = Gas used × Gas price Example: 21000 × 2 Gwei = 42,000 Gwei = 0.000042 ETH (1 Gwei = 10^{-9} ETH)

Conclusion (write this)

MetaMask was installed and configured on the Sepolia test network. Every transaction costs **gas used × gas price**. A simple transfer always uses 21,000 gas, but the gas price changes with network demand. A higher priority fee confirms faster and costs more.